┌Lemma. Let $\varphi: F \to F'$ be a field Isomorphism, and let $\varphi^*: F[x] \to F'[x]: \sum r_i x^i \mapsto \sum \varphi(r_i) x^i$
Let $p(x) \in F[x]$ be irreducible, and let $p^*(x) = \varphi^*(p(x))$.
If β is a root of $p(x)$ and β' is a root of $p^*(x)$,
then, there exists a unique Isomorphism $\overline{\varphi}: F(\beta) \mapsto F'(\beta')$ s.t $\overline{\varphi}|_F = \varphi$, $\overline{\varphi}(\beta) = \beta^*$.

Proof. Since φ^* is also Isomorphism,

$$\Sigma: F[x]/(p(x)) \to F'[x]/(p^*(x)): f(x) + (p(x)) \mapsto \varphi^*(f(x)) + (p^*(x))$$

It maps $c + (p(x)) \mapsto \varphi(c) + (p^*(x))$ for all $c \in F$ and $x + (p(x)) \mapsto x + (p^*(x))$, and Isomorphism.

Now,
$$\overline{\varphi}: F(\beta) \xrightarrow{\phi^{-1}} F[x]/(p(x)) \xrightarrow{\Sigma} F'[x]/(p^*(x)) \xrightarrow{\phi'} F'(\beta')$$
$$: \alpha \mapsto (\phi' \circ \Sigma \circ \phi^{-1})(\alpha)$$
is a field isomorphism leaving F fixed, and β maps to β'.

┌Lemma. Let $f(x) \in F[x]$ and $D_x f(x) \in F[x]$ be a derivative of $f(x)$. Then,

$f(x)$ has a multiple root α if and only if $D_x f(x)$ has α as a root.

In particular,

$f(x)$ is separable if and only if $(f(x), D_x f(x)) = 1$.

Proof. Suppose that $(x-\alpha)^2$ divides $f(x)$ in the splitting field K of f. Then,

$$f(x) = (x-\alpha)^2 \cdot h(x) \quad \text{for some } h(x) \in K[x].$$

By the property of derivative,

$$D_x f(x) = 2 \cdot (x-\alpha) \cdot h(x) + (x-\alpha)^2 \cdot D_x h(x) = (x-\alpha)(2h(x) + (x-\alpha) \cdot D_x h(x)).$$

Thus proved. Conversely, suppose that both f and $D_x f$ has α as a root. Then,

$f(x) = (x-\alpha) \cdot h(x)$ for some $h(x) \in K[x]$ and $D_x f(x) = h(x) + (x-\alpha) \cdot D_x h(x)$.

And, by the assumption, $(x-\alpha)$ divides $D_x f$, so divides $h(x)$. Now,

$$f(x) = (x-\alpha) h(x) = (x-\alpha)^2 r(x), \quad \text{where } r(x) = h(x)/(x-\alpha).$$

Now, the last assertion is obtained by;

f is separable iff f has no multiple root iff $\deg (f(x), D_x f(x)) = 0$ iff $(f(x), D_x f(x)) = 1$.

「Thm. A field of characteristic 0 is perfect.

Proof. Let $f(x) \in F[x]$ be an irreducible polynomial. If $D_x f(x)$ is not zero, then

$$\deg D_x f < \deg f.$$

Therefore, $(f, D_x f) = 1$, because if $(f, D_x f) \neq 1$, then it is contradiction to f is irreducible..
So it is separable.

「Lemma. A field of characteristic p is perfect iff every element of F has a pth root in F.

Proof. Let F be a field of characteristic p.
Suppose that F is perfect. Let $\alpha \in F$ be a non-zero, non-1. Put $f(x) = x^p - \alpha$.
If $f(x)$ has a root in F, then there is nothing to prove.
Assume f has no root in F. Observe that in splitting field K of $f(x)$, put $\beta \in K$ s.t

$$f(\beta) = \beta^p - \alpha = 0.$$

And, $f(x) = x^p - \alpha = x^p - \beta^p = (x-\beta)^p$ in K. But, if f is reducible in F, then for some $1 \leq k < p$, $f(x) = (x-\beta)^{p-k} \cdot (x-\beta)^k$ and $(x-\beta)^{p-k}, (x-\beta)^k \in F[x]$.

That is, $(-\beta)^k \in F$. Clearly, $(k, p) = 1$, thus $ku + pv = 1$ for some $u, v \in \mathbb{Z}$.
Now, $\beta = \beta^{ku+pv} = (\beta^k)^u \cdot (\beta^p)^v = (\beta^k)^u \cdot \alpha^v \in F$. It is contradiction. But, if f is irreducible then $D_x f = p \cdot x^{p-1} = 0$ implies contradiction with F is perfect. Now, α has a p'th root in F.
Conversely, let $f(x) \in F[x]$ be an irreducible. Assume it is inseparable.
Denote $f(x) = a_n x^n + \cdots + a_1 x + a_0$. Since $f(x)$ is irreducible and inseparable,
$D_x f(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \cdots + a_1$ must be zero, that is,

$$a_k \neq 0 \iff k \in (p)$$

Therefore, we can write

$$f(x) = b_m \cdot x^{mp} + b_{m-1} x^{(m-1)p} + \cdots + b_1 x^p + b_0$$

Now, by the assumption, each b_i has p'th root c_i, thus

$$f(x) = c_m^p \cdot (x^m)^p + c_{m-1}^p \cdot (x^{m-1})^p + \cdots + c_0^p$$

$$= (c_m \cdot x^m + c_{m-1} \cdot x^{m-1} + \cdots + c_0)^p.$$ But it is reducible. Contradiction.

「Corollary. Every finite field is perfect. ($a \mapsto a^p$ is 1-1 on finite, thus Auto.)

「Thm. Every finite separable extension is simple extension.

Proof. If F is finite, then the finite extension E/F is also finite,
Then, $E^{\times}$ is cyclic, therefore $E = F(\alpha)$ where α is generator of $E^{\times}$.

Now, assume that F is infinite. Since E/F is finite,

$$E = F(\alpha_1, \cdots, \alpha_r) \quad \text{where } \alpha_1, \cdots, \alpha_r \in E.$$

Now, enough to show $F(\beta, \gamma) = F(\alpha)$ for some $\alpha \in E$, and after is solved by the induction.

Let $\beta_1, \cdots, \beta_n$ be distinct zeros of $\mathrm{irr}(\beta, F)$, and
$\gamma_1, \cdots, \gamma_m$ be distinct zeros of $\mathrm{irr}(\gamma, F)$, in $\overline{F}$. ($\beta_1 = \beta$, $\gamma_1 = \gamma$).

Since E/F is separable, each zeros are multiplicity of 1.

Take $a \in F$ s.t. $a \neq \dfrac{\beta_i - \beta}{\gamma - \gamma_j}$ for all $1 \le i \le n$, $2 \le j \le m$, since F is infinite.

Put $\alpha = \beta + a \cdot \gamma$. Then, for all $(i, j) \in \{1, \cdots, n\} \times \{2, \cdots, m\}$,

$$\alpha = \beta + a\gamma \neq \beta_i + a \cdot \gamma_j, \quad \text{or} \quad \alpha - a \cdot \gamma_j \neq \beta_i.$$

observe that: $\mathrm{irr}(\beta, F)(\alpha - ax) \in F(\alpha)[x]$, and

$$\mathrm{irr}(\beta, F)(\alpha - a\gamma) = \mathrm{irr}(\beta, F)(\beta) = 0.$$

But, $\mathrm{irr}(\beta, F)(\alpha - a \cdot \gamma_i) \neq 0$ for all $2 \le i \le m$,

That is, $\mathrm{irr}(\beta, F)(\alpha - ax)$ and $\mathrm{irr}(\beta, F)(x)$ have a common factor $(x - \gamma)$ in $F(\alpha)$,
thus $\gamma \in F(\alpha)$, so $\beta = \alpha - a\gamma \in F(\alpha)$. ∎